

Online Segment 3D Gaussians via Launching Virtual Drones

Supplementary Material

6 Computational analysis

To further analyze the efficiency of SAGO, we conducted a statistical analysis of the segmentation time and the number of segmentation steps for four datasets including SPIn-NeRF [31], NVOS [36], 3D-OVS [26], and LERF-Mask [16].

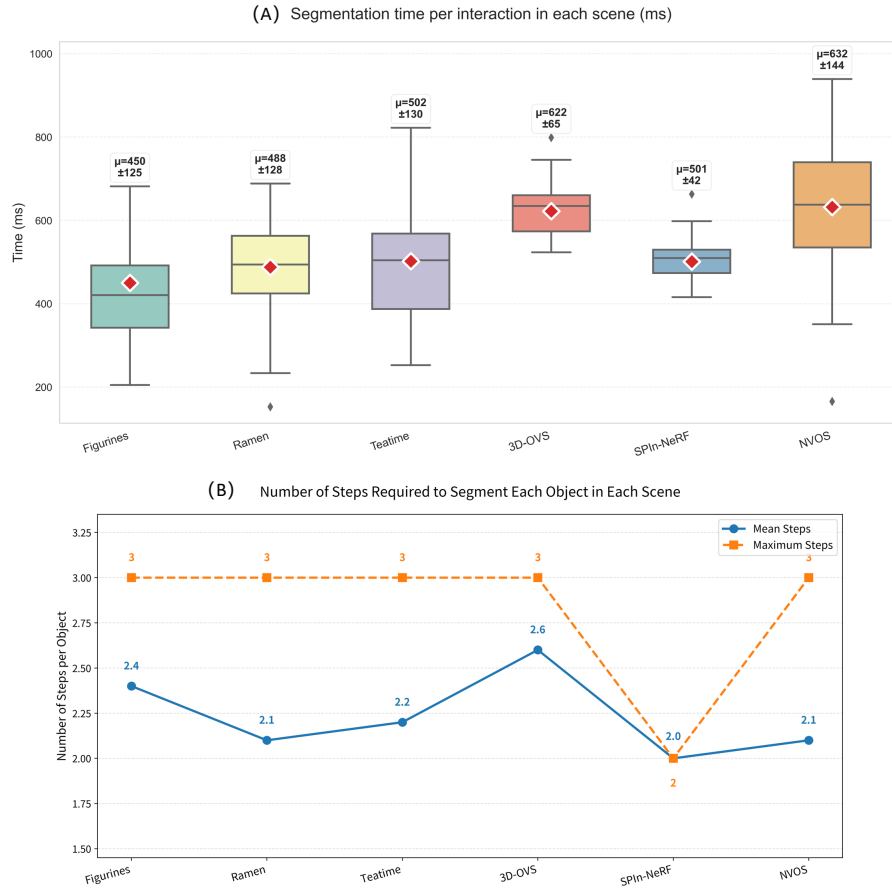


Fig. 9: Computational analysis. (A) Segmentation time per interaction in each scene. (B) Average and maximum step per segmentation. Note that ‘Figurines’, ‘Ramen’ and ‘Teatime’ are from LERF-Mask dataset. Due to the complexity of these three scenes, we provide a separate analysis.

6.1 Inference Time Analysis

As shown in Fig. 9 (A), our statistical analysis across multiple datasets indicates that the segmentation time per interaction ranges from 0.3 to 1 second, with an average of approximately 0.5 seconds. The results demonstrate that even without the setup stage, our proposed method exhibits strong stability in terms of interactive segmentation latency. The segmentation time we report does not include the initial mask time generated during user interaction, which typically takes 0.2-0.5 seconds. We consider this time as part of the interaction time, and we also exclude this time when comparing with other methods.

6.2 Markov State Transfer Count Analysis

As described in Sec. 3.4, in the worst case the yaw step size is limited to 30° , requiring up to 6 Markov states ($180^\circ/30^\circ = 6$) to achieve the full angular coverage needed for segmentation. Fortunately, as shown in Fig. 9 (B), thanks to the carefully designed NBV planning strategy, experiments on four datasets [16, 26, 31, 36] show that no scene required more than three Markov states to complete the segmentation.

Theoretical minimum state count.

When counting the number of states, we exclude the initial state S_0 . As shown in Fig. 10, our bidirectional NBV implies that the theoretical minimum number of Markov states is **2** (S_1 and S_2). In practice, our experiments show that most scenes can be fully segmented using only 2-3 Markov states, demonstrating the effectiveness of our NBV planning strategy. Our NBV planning for the virtual drone prioritizes paths with the minimum state count. It only explores paths with smaller yaw spans when an

excessively large span causes the rendered texture of the target object to become overly large, thereby degrading SAM2’s tracking performance. In addition to relying on SAM2 tracking, our method for acquiring new-view masks also incorporates the 3D position of the target object as an additional spatial constraint to assist segmentation. This is one of the reasons why our method can complete segmentation in most scenes using only 2-3 Markov states.

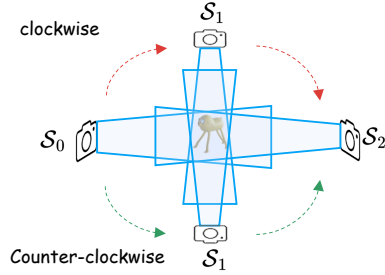


Fig. 10: Theoretical minimum state count. We use bidirectional NBV planning, so the theoretical minimum number of Markov states is **2** (i.e., S_1 and S_2).

7 Implementation details

mIOU threshold. In our implementation, we set the σ threshold in Eq. (6) to 0.95. Our guiding principle is that segmentation from a new viewpoint must not

degrade the rendering quality of the initial viewpoint. We use Eq. (6) to identify a suitable viewpoint at which SAM2 can track the target effectively. If the yaw span is too small, more Markov states are required, which significantly increases the segmentation time. Therefore, our proposed method first explores viewpoints with a large yaw span (90°). Only if the tracking performance is poor (first-view $\text{mIoU} < \sigma$) does it then explore smaller yaw spans. As shown in Fig. 9, the maximum number of steps required for segmentation is 3, which demonstrates the efficiency of our NBV planning strategy.

Mask Expansion Coefficient for MFF. In the implementation of Mask-shaped Frustum Filtering (MFF), we apply a mask expansion operation to the segmentation mask obtained from SAM2 before performing the filtering. We apply a mask expansion operation with a mask expansion coefficient e to the binary mask obtained from SAM2 before performing MFF. In our experiments, we set the mask expansion coefficient e to 3.

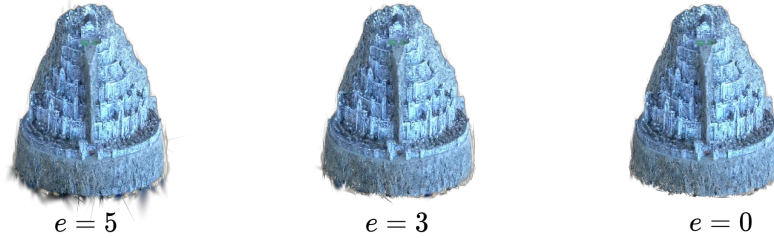


Fig. 11: Ablation study on Mask Expansion. The three mask expansion coefficients (0, 3, and 5) are applied to the segmentation mask obtained from SAM2 before performing MFF.

As shown in Fig. 11, we conducted an ablation study on the mask expansion coefficient for MFF. We tested three different coefficients (0, 3, and 5) applied to the segmentation mask obtained from SAM2 before performing MFF. From a practical perspective, there is no optimal mask expansion coefficient. This serves as a user option: if you want a more complete segmentation result that may have edge artifacts, you can choose a larger mask expansion coefficient; if you want sharper edges but may lose edge details, you can choose a smaller mask expansion coefficient.

7.1 Anti-occlusion Mechanism

In 3D scenes, mutual occlusions inevitably exist between multiple viewpoints. Prior works have devoted substantial effort to handling occlusions. However, our method can actively eliminate potential occlusions through the Markov process of online NBV planning. This gives our approach a significant advantage in dealing with complex scenes. As shown in Fig. 12, at time t , our virtual drone

can use MFF to clear occlusions for the viewpoint at time $t + 1$, ensuring that the viewpoint at time $t + 1$ can fully cover the target object, thereby achieving complete segmentation.

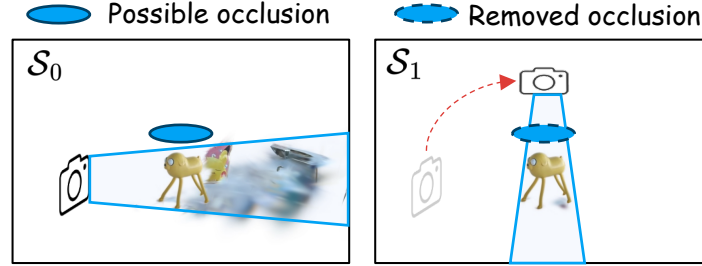


Fig. 12: Illustration of anti-occlusion mechanism. Virtual drones can actively eliminate potential occlusions through the Markov process of online NBV planning.

7.2 GUI-based Implementation for SAGO

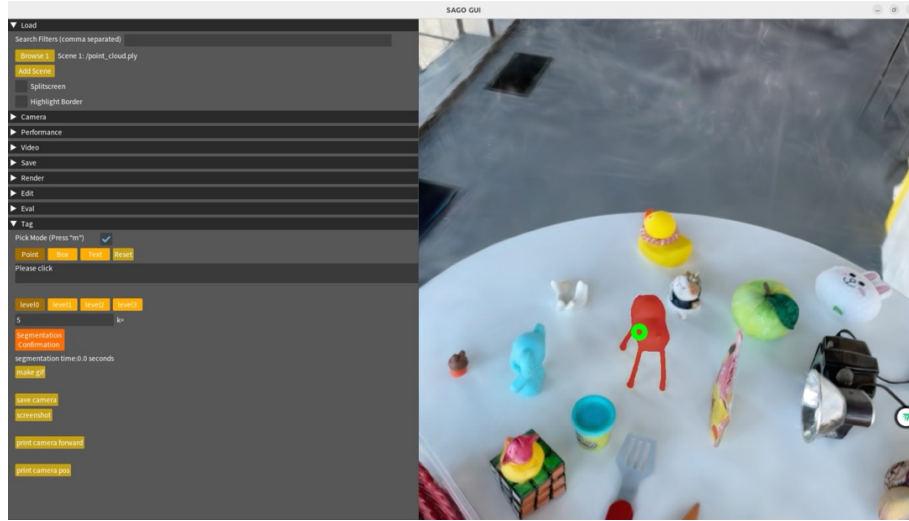


Fig. 13: Illustration of GUI-based implementation for SAGO. The GUI allows users to interactively perform segmentation in 3D scenes, select target objects, visualize segmentation results, and adjust parameters such as the mask expansion coefficient for MFF.

To facilitate the practical application of SAGO, we have developed a GUI-based implementation that allows users to interactively perform segmentation in 3D

scenes. The GUI provides an intuitive interface for users to select target objects, visualize segmentation results, and adjust parameters such as the mask expansion coefficient for MFF. We design this GUI based on SplatViz [2].

8 Additional Qualitative Results

8.1 Online Free Granularity Control

The segmentation granularity of the interactive 3D segmentation results in our method can be completely determined by the 2D segmentation granularity. Users only need to select their preferred 2D segmentation granularity through simple interactive operations such as clicking or drawing bounding boxes, and this 2D granularity will be directly reflected in the 3D segmentation results.

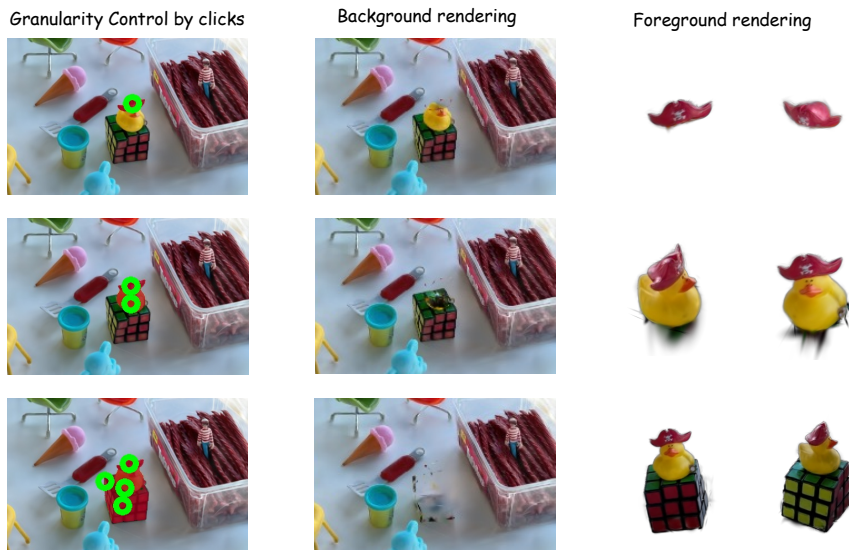


Fig. 14: Illustration of online free granularity control. The 2D granularity selected by the user is directly reflected in the 3D segmentation results.

As shown in Fig. 14, we demonstrate the online free granularity control capability of our method. By adjusting the 2D segmentation granularity through user interactions, users can achieve their desired level of detail in the 3D segmentation results, providing a flexible and user-friendly experience.

8.2 Comparison with Other Online Methods

We conduct a qualitative comparison of our method with other online methods for interactive 3D segmentation. As shown in Fig. 15, our method achieves superior segmentation quality and efficiency compared to other online methods, demonstrating its effectiveness in interactive 3D segmentation tasks.

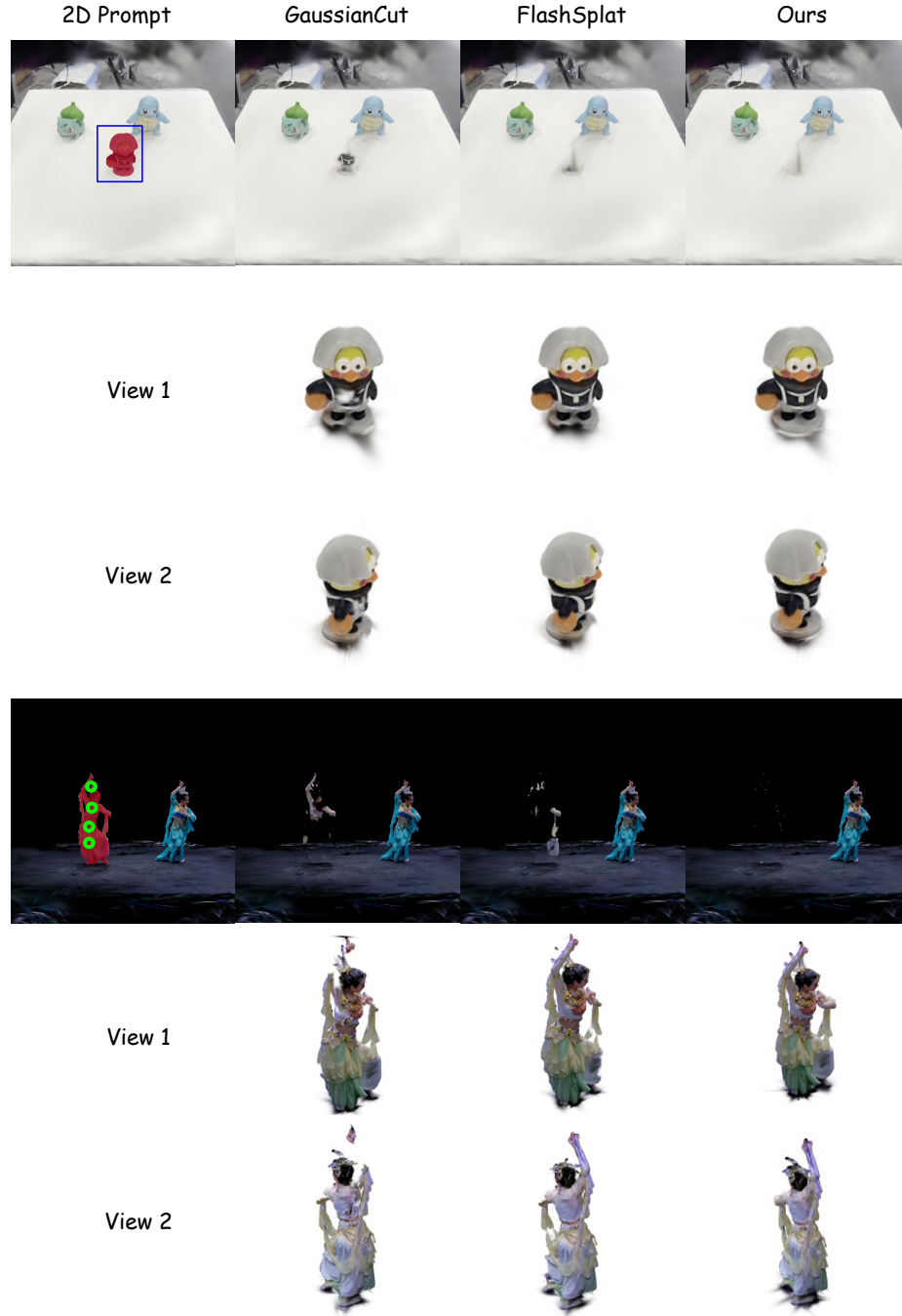


Fig.15: Qualitative comparison with other online methods [13, 38]. Our method achieves superior segmentation quality and efficiency compared to other online methods, demonstrating its effectiveness in interactive 3D segmentation tasks.

9 Limitations and Failure Cases

Dependence on good initial segmentation. Our method relies on the initial segmentation results obtained from user interactions. If the initial segmentation is of poor quality, it may lead to suboptimal segmentation results in subsequent steps. Therefore, our method requires a high-quality initial interactive segmentation result to ensure the effectiveness of the overall segmentation process.

No boundary trimming. The current implementation of our method does not include a boundary trimming mechanism, which may lead to boundary artifacts in the segmentation results. This is because our Mask-shaped Frustum Filtering (MFF) operation is center-based, and since Gaussians have volume, especially at the edges where they often span across two objects, a purely center-based approach cannot effectively address edge artifacts. However, we are currently exploring the integration of a post-processing tool called GaussianTrimmer [25] to mitigate edge artifacts and improve the overall quality of the segmentation results.